Implementation of Shop Floor Management Cycle (SFMC) at BOSCH Ltd., Nashik Plant

Internship Report

Submitted by:

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Department: Body Department (MFC2)

Internship Period: 2 June 2025 – 31 November 2025

November 16, 2025

CERTIFICATE

This is to certify that the **"Semester Long Internship Program / Industry Practicum" Report**, submitted by **Mr. Pradyumna Kulkarni (PRN: 202201090099)**, is a record of work carried out by him during Academic Year 2025–2026 at BOSCH Ltd., Nashik Plant.

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This is to certify that **Mr. Pradyumna Kulkarni** of MIT Academy of Engineering, pursuing B. E Mechanical Engineering was engaged as Project Trainee at Bosch Ltd, Nashik from 02.06.2025 to 30.11.2025 under the guidance of Mr. Sachin Bhandare – Manager, Manufacturing. During his internship, he has worked on the below Project–

- **Implementation of Shop Floor Management Cycle (SFMC) for Imported MAE bottleneck area & data analysis.**

We wish him success in his future endeavors.

For Bosch Limited,

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Acknowledgement

I would like to express my deep sense of gratitude to BOSCH Ltd., Nashik Plant, for granting me the opportunity to undertake my six-month internship in the Body (MFC2) Department. Being associated with one of the most reputed and structured manufacturing organizations has been a truly enriching and transformative experience in my academic journey.

I extend my heartfelt thanks to my company mentor, Mr. Sachin Bhandare, for his constant encouragement, patient guidance, and invaluable insights throughout the project on Shop Floor Management Cycle (SFMC). His ability to explain complex industrial practices in a simple and practical manner helped me develop clarity and confidence in applying engineering principles effectively on the shop floor. His mentorship has been instrumental in shaping my technical understanding and professional attitude.

I am also extremely grateful to my college mentor, R. A. Patil sir, for his continuous support, constructive feedback, and motivation during the course of the internship. His guidance helped me maintain academic discipline, improve documentation, and stay aligned with the learning objectives of the program.

I extend my sincere appreciation to all the engineers, line leaders, supervisors, and operators in the MFC2 department who welcomed me warmly and supported me throughout my learning journey. Their willingness to share knowledge, clarify doubts, and involve me in real-time production and problem-solving activities contributed immensely to my practical learning. Observing their dedication, teamwork, and professionalism has been both inspiring and educational.

A special thanks to the Quality, Maintenance, and Planning teams for offering their time and expertise whenever required. Their combined efforts helped me understand the interdependencies of different functions and the importance of cross-functional coordination in achieving production excellence.

I am grateful to the entire shop floor team whose cooperation enabled me to conduct studies, collect data, analyze bottlenecks, and implement SFMC tools effectively. Their practical experience and insights helped me bridge the gap between theoretical knowledge and real-world applications.

This internship has been a memorable chapter in my academic life, and I feel fortunate to have learned and grown under the guidance of such dedicated professionals. I am

sincerely thankful to everyone who played a part in making this experience meaningful and successful.

Date: November 16, 2025

Executive Summary

The six-month internship at BOSCH Ltd., Nashik Plant, carried out in the Body Department (MFC2), focused on the development and implementation of the Shop Floor Management Cycle (SFMC) to improve production visibility, streamline daily monitoring, and strengthen continuous improvement practices. The project provided hands-on exposure to real manufacturing challenges and an opportunity to directly contribute to operational excellence on the shop floor.

Objectives

- To develop a structured Shop Floor Management Cycle tailored for the MFC2 machining line.
- To design and standardize visual management tools such as hourly monitoring sheets, takt boards, OEE dashboards, and Pareto charts.
- To enhance real-time monitoring of production performance and enable faster escalation of issues.
- To identify bottlenecks using cycle-time and takt-time studies and support improvement actions.
- To strengthen problem-solving capabilities using PSS, 5-Why analysis, and OPL-based operator training.
- To promote accountability and ownership among operators, line leaders, and engineers.

Approach

- Conducted a baseline study of the existing production system, including documentation practices, workflow, cycle times, and past quality data.
- Created a detailed process flow map and performed cycle time vs. takt time comparisons to identify high-variance stations.
- Developed and implemented visual tools such as takt boards, hourly sheets, OEE dashboards, and loss Pareto charts to improve transparency.
- Introduced a tiered meeting structure (Tier 1, Tier 2, Tier 3) to improve communication, escalation, and closure of issues.

- Used structured problem-solving tools such as PSS and 5-Why analysis to address recurring issues.
- Executed improvement actions involving layout changes, standardization of formats, and operator-focused One Point Lessons (OPLs).

Key Outcomes

- Significant improvement in daily production stability and consistency with better adherence to target quantities.
- Reduction in scrap and quality losses through structured tracking and Pareto-based prioritization.
- Decrease in cycle time losses and smoother workflow due to controlled bottlenecks.
- Enhanced operator participation through visual boards, OPL training, and clear KPI ownership.
- Faster and more reliable problem escalation achieved through the tiered meeting structure.
- Improved overall OEE trends driven by reduction in organizational, changeover, technical, performance, and quality losses.
- Strengthened technical skills, including KPI interpretation, data analysis, lean tools, and process understanding.

This structured SFMC implementation has provided a sustainable framework for daily management and has significantly improved shop-floor performance. The internship served as an enriching learning experience, offering both technical and professional growth while contributing to the continuous improvement culture of BOSCH Ltd., Nashik Plant.

Contents

Company Certificate i

Acknowledgement ii

Executive Summary iv

1 Introduction 1

1.1 Internship Background ... 1
1.2 About BOSCH Ltd. ... 1
1.3 About the Department (MFC2) ... 2
1.4 Internship Objectives ... 2

2 Technical Background 4

2.1 Shop Floor Management (SFM) ... 4
2.2 Shop Floor Management Cycle (SFMC) ... 4
2.3 Key Performance Indicators (KPIs) in SFMC ... 4
2.4 Takt Time and Cycle Time Concepts ... 7
2.5 Overview of Fuel Injector Body Manufacturing (Relevant to MFC2) ... 8
2.6 Data Sources and Measurement Systems ... 8

3 SFMC Project: Implementation and Study 9

3.1 Project Overview ... 9
3.1.1 Title ... 9
3.1.2 Background ... 9
3.1.3 Problem Statement ... 9
3.1.4 Objectives ... 9
3.2 Existing Situation Analysis ... 9
3.3 Process Flow Study ... 10
3.4 Development of SFMC Structure ... 11
3.4.1 Concept and Design Philosophy ... 11
3.4.2 Visualization Tools Implemented ... 11
3.5 Tiered Meeting Structure ... 11
3.5.1 Tier 1 ... 11

3.5.2 Tier 2 11
3.5.3 Tier 3 11
3.6 Bottleneck Analysis 11
3.7 Improvement Actions and Kaizens 12
3.8 Results and Impact 12
3.8.1 Quantity Improvements 12
3.8.2 Quality Improvements 12
3.8.3 Productivity Improvements 13
3.9 Summary 13

4 Major Outcomes and Learnings 14
4.1 Major Outcomes 14
4.1.1 Improved Production Stability 14
4.1.2 Enhanced Quality Control 14
4.1.3 Strengthened Problem-Solving Culture 14
4.1.4 Bottleneck Reduction 15
4.1.5 Improved Operator Engagement 15
4.1.6 Increase in OEE 15
4.2 Technical Learnings 15
4.2.1 Understanding of SFMC Framework 15
4.2.2 Proficiency in Lean Manufacturing Tools 15
4.2.3 Data Analysis and Interpretation 15
4.2.4 Root Cause Analysis 16
4.2.5 Process Understanding 16
4.3 Soft Skills and Professional Development 16
4.3.1 Communication 16
4.3.2 Teamwork and Collaboration 16
4.3.3 Problem-Solving Mindset 16
4.3.4 Time Management and Ownership 17
4.4 Summary 17

5 Conclusion 18

A Annexures 20
A.1 Sample Hourly Monitoring Sheet 20
A.2 PSS and OPL Templates 20
A.3 Additional Figures and Charts 20

List of Figures

2.1 Example of loss-wise Pareto analysis (placeholder). 6

3.1 Process flow diagram of the MFC2 machining line (placeholder). 10

A.1 Hourly monitoring sheet (sample) 20

List of Tables

2.1 Sample definitions and relation 7

3.1 Summary of major improvements implemented 12

1. Introduction

1.1 Internship Background

This report documents a six-month industrial internship undertaken at BOSCH Ltd., Nashik Plant, during the period 2 June 2025 – 31 November 2025. The internship was carried out in the Body Department (MFC2) as part of the requirement for the Bachelor of Technology degree in Mechanical Engineering at MIT Academy of Engineering. The primary focus of the internship was the planning, development and implementation of a Shop Floor Management Cycle (SFMC) to improve visibility of production performance, standardize daily review practices, and reduce bottlenecks in the shop-floor processes.

During the internship I was involved in data collection, takt-time and cycle-time studies, development of visualization tools (hourly monitoring sheets, takt boards, OEE dashboards), facilitation of tiered review meetings, and execution of root-cause analysis and countermeasures using standard lean problem-solving tools (Pareto, 5-Why, PDCA). Work was performed under the guidance of the company mentor, Mr. Sachin Bhandare, and the college guide, R. A. Patil.

1.2 About BOSCH Ltd.

Bosch is a multinational engineering and technology company founded in 1886. The Nashik facility specializes in manufacturing precision components for automotive fuel injection systems and related products. The plant combines machining, assembly and testing operations across multiple production lines and supports a large workforce operating under strict quality and safety norms. The plant's strong process-control culture, documented operating procedures and established continuous-improvement practices provided an ideal environment for implementing a structured SFMC system.

Key contextual details relevant to this internship include:

- **Product focus:** Fuel injector bodies and related subassemblies requiring precision machining and stringent quality control.
- **Production environment:** Multiple machining stations and assembly cells arranged in process flows with defined takt times and operator responsibilities.

- **Quality & metrics:** Emphasis on KPIs such as OEE, rejection rate, cycle time, and first-pass yield for daily performance monitoring and decision-making.

1.3 About the Department (MFC2)

The Body Department (MFC2) handles critical machining and pre-assembly operations for injector bodies. Typical activities include CNC turning, milling, drilling, gauging, and post-process inspection. The department operates several production cells, each with its own set of machines, fixtures and operator teams. Production planning, tool maintenance, and inline inspection are key supports to ensure that dimensional and functional requirements are met.

Prior to the SFMC implementation, the department faced challenges in standardized information flow, inconsistent hourly recording of performance, and unclear escalation paths for recurring issues. These gaps affected the ability of supervisors and engineers to react in real time and to prioritize improvement activities effectively.

1.4 Internship Objectives

The internship was structured to provide both technical exposure and practical involvement in live manufacturing operations at BOSCH Ltd., Nashik Plant. Rather than simply observing production activities, the internship focused on applying engineering principles, lean tools, and structured problem-solving methods to support the implementation of the Shop Floor Management Cycle (SFMC). The detailed objectives of the internship are as follows:

1. **Design and implement a complete SFMC framework:** Establish a structured daily management system that includes tier meetings, KPI review mechanisms, issue escalation flow, and visual performance tracking.
2. **Enhance real-time process visibility:** Develop visualization tools such as hourly monitoring sheets, takt boards, rejection dashboards, breakdown logs, and OEE displays.
3. **Conduct systematic bottleneck identification:** Perform cycle time studies, compare takt time versus actual production time, and analyze downtime and scrap trends using Pareto techniques.
4. **Strengthen standardized problem-solving practices:** Use PSS (Problem Solving Sheet), 5-Why analysis, Ishikawa diagrams, and OPLs (One Point Lessons) to address recurring issues.
5. **Improve accountability and ownership on the shop floor:** Encourage operators and line leaders to track their performance through visual KPIs, participate actively in daily meetings, and take responsibility for action closure.

6. **Support productivity, quality, and cost optimization:** Work on reducing scrap, minimizing cycle losses, improving machine availability, and supporting smooth process flow.
7. **Develop engineering and managerial competencies:** Enhance technical skills in data handling, chart preparation, process analysis, and documentation, along with soft skills including communication, teamwork and coordination.

The above objectives ensured that the internship provided holistic exposure—not only in manufacturing technology but also in professional problem-solving, leadership, and industrial decision-making.

This introduction sets the context for the technical background, methodology, and detailed project activities that follow in the subsequent chapters.

2. Technical Background

2.1 Shop Floor Management (SFM)

Shop Floor Management (SFM) is a structured management approach focused on ensuring that production activities on the shop floor are controlled, transparent, and continuously improving. The primary goals of SFM are to:

- Ensure real-time visibility of production performance and quality.
- Provide a structured forum for daily communication between operators, supervisors, and engineers.
- Enable immediate containment and systematic root cause elimination of problems.
- Drive continuous improvement through short PDCA cycles and operator-led Kaizens.

2.2 Shop Floor Management Cycle (SFMC)

SFMC organizes daily shop-floor activities into a repeating loop of observation, review, escalation, improvement and verification. A typical SFMC includes:

1. Visual performance boards and dashboards.
2. Hourly data capture and status indication.
3. Tiered meetings for escalation and decision-making.
4. Structured problem-solving (PSS, 5-Why, Ishikawa).
5. Verification of effectiveness and standardization.

Tier Meeting Structure

Tier 1: Operator and Line Leader — immediate containment and quick corrective actions.
Tier 2: Line Leader and Area Manager — deeper analysis and resource alignment.
Tier 3: Department Head — systemic improvements and major decisions.

2.3 Key Performance Indicators (KPIs) in SFMC

KPIs are essential to the SFMC because they provide objective data for decision-making and escalation. Typical KPIs used in MFC2 included:

- **Overall Equipment Effectiveness (OEE) —**

In the MFC2 department of BOSCH Ltd., Nashik Plant, OEE is calculated using a simplified and practical productivity-based formula. Instead of breaking OEE into Availability, Performance, and Quality percentages, the department directly compares the actual output with the planned output for the day.

$$\text{OEE} = \frac{\text{Total Parts Produced per Day}}{\text{Target Quantity}} \tag{2.1}$$

Example: If 870 parts are produced against a target of 950:

$$\text{OEE} = \frac{870}{950} = 0.915\ (91.5\%)$$

OEE Loss Categories

1. **Quality Loss:**

$$\text{Quality Loss} = \frac{\text{Rejected Parts}}{\text{Total Parts Produced}}$$

2. **Organizational Loss:** Loss of production due to non-availability of:
   - Operators
   - Materials
   - Tools
   - Waiting for instructions
3. **Changeover Loss:** Time lost due to setup changes, tool change, fixture change, or product variant switching.
4. **Technical Loss:** Machine breakdowns, tool failure, alarms.
5. **Performance Loss:** Running below expected cycle time or micro stoppages.

These losses are added and compared against the total available production time to identify improvement areas. A Pareto analysis is typically used to prioritize the top contributors.

figures/pareto_loss_example.png

Figure 2.1: Example of loss-wise Pareto analysis (placeholder).

Cycle Time / Takt Time adherence — measures how actual cycle times compare with takt requirements.

Rejection Rate / Scrap (%) — fraction of produced parts failing inspection.

First Pass Yield (FPY) — proportion of parts passing all checks first time.

Downtime (and its causes) — planned vs unplanned and categorized by root cause.

Below are the primary lean and quality tools applied during the SFMC implementation, with short descriptions and how they were used in the project.

5S

A workplace organisation method: Sort, Set in order, Shine, Standardize and Sustain. At MFC2, 5S improved material flow, reduced search time and helped maintain consistent visual standards on operator work-stations.

PDCA (Plan-Do-Check-Act)

Used for iterative improvements — plan interventions (e.g., change hourly sheet layout), pilot them on a small scale, check results (data from hourly monitoring), and act (standardize

or iterate).

5-Why and Ishikawa (Fishbone)

Structured root-cause techniques used during PSS preparation to trace problems back to causal factors (man, machine, method, material, measurement, environment).

OPL (One-Point Lessons) and Standard Work

OPLs were prepared for critical operator tasks and quick troubleshooting steps — they support training and reduce variation in execution.

Problem Solving Sheet (PSS)

Standardized PSS format was implemented to capture problem description, containment, root cause analysis, corrective / preventive action and verification results. This ensured traceability and closure.

2.4 Takt Time and Cycle Time Concepts

Takt time is the rate at which a product must be completed to meet customer demand. It is defined as:

$$\text{Takt Time} = \frac{\text{Available Production Time}}{\text{Customer Demand (units)}}$$

Cycle time refers to the actual processing time of a part at a station. A station becomes a bottleneck when its cycle time exceeds takt time or when effective output lags its required throughput.

Table 2.1: Sample definitions and relation

Term	Meaning / Use
Takt Time	Customer-driven pace target for production.
Cycle Time	Actual time to complete a unit at a station.
Bottleneck	Station limiting throughput; needs priority for improvement.
OEE	Aggregate measure used to compare productivity across machines/lines.

2.5 Overview of Fuel Injector Body Manufacturing (Relevant to MFC2)

This subsection provides a condensed process description focusing on stages relevant to the body department.

1. **Raw Material Machining:** Bars/forgings are mounted on CNC turning/milling centers to produce basic geometry.
2. **Drilling & Boring:** Critical internal features are machined and gauged for dimensional accuracy.
3. **Grinding (if required):** To achieve surface finish and critical tolerances where functional sealing or fit is needed.
4. **Deburring & Cleaning:** To remove machining burrs and prepare parts for assembly.
5. **Inspection and Gauging:** In-process gauging ensures features remain within tolerance; key dimensions are checked before release to assembly.
6. **Assembly Preparation:** Parts are staged with fixtures, OPLs and visual checks to ensure smooth downstream assembly.

Critical process considerations:

- Dimensional tolerances and surface finish directly affect sealing and functional performance of injectors.
- Tool wear, machine thermal drift and fixturing issues are common contributors to variation.
- Inline gauging and quick feedback loops (via SFMC) help detect and contain drift before large batches are affected.

2.6 Data Sources and Measurement Systems

Reliable data capture is central to SFMC. During this internship the following sources were used:

- Hourly monitoring sheets filled on the shop floor (manually / semi-digital).
- Machine logs from CNC controllers (cycle counts, alarm logs).
- Quality inspection records (gauge readings, SPC charts).
- Maintenance records (CLIT charts[Cleaning, Lubrication, Inspection, Tightining],breakdown times, corrective actions).

Measurement accuracy and calibration: Measurement devices (gauges, micrometers) were checked for calibration status before using their readings for critical decisions. Any suspect measurement trend was validated via repeat checks or alternate instruments.

3. SFMC Project: Implementation and Study

3.1 Project Overview

3.1.1 Title

Implementation of the Shop Floor Management Cycle (SFMC) in the Body Department (MFC2) at BOSCH Ltd., Nashik Plant.

3.1.2 Background

Initial observations found:

- Inconsistent monitoring of hourly production.
- Limited visibility of real-time KPIs.
- Delayed escalation of quality/machine issues.
- Lack of documented root-cause countermeasures.

3.1.3 Problem Statement

Major challenges before SFMC:

- No unified visual management system.
- Absence of tier-based daily review.
- Production instability due to bottlenecks.

3.1.4 Objectives

- Establish a structured and sustainable SFMC system.
- Standardize visual performance tracking.
- Reduce bottlenecks using cycle/takt time analysis.
- Improve OEE and operator accountability.

3.2 Existing Situation Analysis

Key findings:

- No standard hourly sheet format.

- No strategy meeting discussions

3.3 Process Flow Study

A detailed process flow for the injector body machining line was created to map all stations, cycle times, inspections, and material movements. The study included:

- Collection of cycle time data from each workstation.
- Comparison of actual cycle times with takt time.
- Identification of high-variance stations.
- Understanding the dependency of upstream and downstream processes.

A sample process flow diagram is shown in Figure 3.1.

figures/process_flow.png

Figure 3.1: Process flow diagram of the MFC2 machining line (placeholder).

3.4 Development of SFMC Structure

3.4.1 Concept and Design Philosophy

The SFMC framework was designed to support real-time monitoring, quick escalation, and structured problem-solving. The design philosophy emphasized:

- Visual management at every workstation.
- Data-driven decision making.
- Operator ownership of daily performance.
- Quick containment and long-term correction.

3.4.2 Visualization Tools Implemented

A complete set of visual tools were developed, tested, and deployed on the shop floor:

- **Takt Board:** Displays planned vs. actual output.
- **Hourly Monitoring Sheets:** Manual sheets updated hourly by operators.
- **OEE Dashboard:** Tracks availability, performance, and quality.
- **Attendance and Manpower Allocation Sheet:** Shows operator assignment.
- **PSS Board:** Documents problem-solving activities.
- **OPL Board:** Provides quick lessons for standard work and training.
- **Pareto Charts:** Display scrap and downtime trends.
- **PDCA Cycle sheet**

3.5 Tiered Meeting Structure

3.5.1 Tier 1

Operators + Line Leader(FLM): immediate actions.

3.5.2 Tier 2

Line leader(FLM) + Area Manager(Planner): resource planning.

3.5.3 Tier 3

Department Head + CFT(Cross Functional Team): systemic decisions.

3.6 Bottleneck Analysis

- Cycle time vs takt comparisons.
- Pareto of downtime/scrap.

3.7 Improvement Actions and Kaizens

Following the identification of major issues, improvement actions were undertaken. Examples include:

- Standardization of OEE tracking procedure.
- Revision of machine cleaning and inspection checklist.
- Introduction of OPL sessions focusing on recurring issues.
- Creation of escalation matrix for Tier meetings.
- Rearrangement of visual boards to improve accessibility.

A summary of the improvement actions is provided in Table 3.1.

Table 3.1: Summary of major improvements implemented

Improvement Area	Description
OEE Dashboard	Developed unified format combining availability, performance, quality.
OPL Training	Conducted quick training sessions to address recurring mistakes.
PSS Documentation	Introduced structured format for root-cause analysis.
Visual Layout Optimization	Reorganized boards for better operator visibility.

3.8 Results and Impact

The implementation of the SFMC framework brought measurable improvements across production, quality and accountability.

3.8.1 Quantity Improvements

- Consistent achievement of daily target quantity.
- Improved adherence to takt time.
- Smoother production flow due to reduced bottlenecks.

3.8.2 Quality Improvements

- Significant reduction in scrap percentage.
- Better traceability and prevention of repeated defects.
- Improved inspection discipline.

3.8.3 Productivity Improvements

- Reduced cycle time losses.
- Faster problem escalation and closure.
- Increased operator involvement and ownership.

3.9 Summary

The SFMC implementation created a more structured, transparent and data-driven environment in the MFC2 department. Operators gained ownership, engineers received better insights, and the overall efficiency of the line improved. The systematic approach introduced during the internship provides a strong foundation for future enhancements, digitalization, and further lean initiatives.

4. Major Outcomes and Learnings

This chapter summarizes the technical, analytical, and professional competencies developed during the internship at BOSCH Ltd., Nashik Plant. The implementation of the Shop Floor Management Cycle (SFMC) not only strengthened process visibility and performance monitoring but also contributed significantly to personal and technical growth.

4.1 Major Outcomes

4.1.1 Improved Production Stability

The SFMC system enabled consistent monitoring of hourly output, machine conditions, and manpower availability. As a result:

- Daily production targets became more predictable and consistent.
- Deviations were detected earlier, reducing risks of end-of-shift backlogs.
- Alignment between planned vs. actual production improved.

4.1.2 Enhanced Quality Control

Structured tracking of scrap and rejection categories through Pareto charts and PSS documentation led to:

- Reduction in dominant defect modes.
- Faster containment of quality issues using Tier 1 actions.
- Better operator awareness regarding defect triggers and prevention.

4.1.3 Strengthened Problem-Solving Culture

With the introduction of PSS and 5-Why analysis:

- Systematic root-cause identification became a standard practice.
- Corrective and preventive actions were defined clearly.
- Follow-ups ensured actions were verified and sustained.

4.1.4 Bottleneck Reduction

The detailed cycle-time and takt-time study enabled the identification of several bottleneck stations. Following improvements:

- Cycle-time variation reduced across stations.
- Machine idle time and micro-stoppages decreased.
- Process synchronization improved, enhancing line balance.

4.1.5 Improved Operator Engagement

Visual boards, hourly sheets, and OPLs fostered operator involvement:

- Operators developed ownership of KPI performance.
- Communication between operators and line leaders improved.
- Tier meetings encouraged active feedback and suggestions.

4.1.6 Increase in OEE

By improving availability, performance, and quality:

- Machine uptime increased due to timely escalation of issues.
- Performance losses reduced as monitoring became more disciplined.
- Rejection-related quality losses decreased.

4.2 Technical Learnings

4.2.1 Understanding of SFMC Framework

Learned how a Shop Floor Management Cycle integrates daily monitoring, visualization, tiered communication, and problem-solving into a unified system.

4.2.2 Proficiency in Lean Manufacturing Tools

- 5S and workplace organization.
- PDCA cycle for iterative improvements.
- Pareto analysis for prioritizing critical issues.
- SPC concepts and use of basic quality control tools.
- OEE breakdown and calculation methodology.

4.2.3 Data Analysis and Interpretation

Gained experience in:

- Collecting and validating production data.

- Identifying trends in scrap, downtime, and cycle time.
- Using graphical tools (bar charts, Pareto charts, run charts).
- Preparing structured daily and weekly reports.

4.2.4 Root Cause Analysis

Improved competency in:

- Constructing Ishikawa (Fishbone) diagrams.
- Conducting 5-Why analysis.
- Preparing comprehensive PSS documentation.

4.2.5 Process Understanding

Deep insights were gained into:

- Injector body machining sequence and tolerances.
- Critical dimensions and inspection requirements.
- Machine-tool behavior and tool wear effects.

4.3 Soft Skills and Professional Development

4.3.1 Communication

The daily interaction with operators, supervisors, and engineers strengthened the ability to:

- Communicate technical issues clearly.
- Participate in structured meetings.
- Document actions and share findings effectively.

4.3.2 Teamwork and Collaboration

Through cross-functional coordination:

- Learned to collaborate with Quality, Maintenance, and Planning teams.
- Understood interdependencies across functions.
- Developed respect for diverse roles in manufacturing.

4.3.3 Problem-Solving Mindset

Exposure to live production problems improved:

- Analytical thinking.
- Decision-making under time constraints.
- Ability to approach problems structurally rather than reactively.

4.3.4 Time Management and Ownership

Handling daily KPIs, reporting, and Tier meetings taught:

- Prioritization of tasks.
- Consistency in daily responsibilities.
- Ownership of assigned improvement activities.

4.4 Summary

Overall, the internship offered a balance of technical exposure, analytical learning, and professional development. The implementation of SFMC served as a strong foundation to understand real-world manufacturing challenges and the importance of lean methodologies. The learnings gained during this period contribute significantly to future roles in manufacturing, operations, and industrial engineering.

5. Conclusion

The six-month internship at BOSCH Ltd., Nashik Plant provided an invaluable opportunity to apply academic knowledge to real industrial practices and understand the functioning of a world-class manufacturing environment. Through involvement in the implementation of the Shop Floor Management Cycle (SFMC) in the Body Department (MFC2), practical exposure was gained in lean manufacturing, structured problem-solving, and daily performance management.

The project enabled close interaction with operators, line leaders, engineers, and cross-functional teams, providing a holistic understanding of production, quality, maintenance, and planning activities. The development and deployment of SFMC tools—including hourly monitoring sheets, takt boards, OEE dashboards, PSS formats, and visual management layouts—helped transform the shop floor into a more transparent, data-driven, and disciplined workspace.

The systematic approach enabled by tier meetings and problem-solving frameworks strengthened communication and improved both short-term and long-term decision-making. Identification and reduction of bottlenecks, improvements in scrap trends, and increased operator engagement demonstrated the effectiveness of structured shop floor management. The internship significantly strengthened technical skills, analytical abilities, and professional competencies. It enhanced understanding of KPI interpretation, cycle-time studies, lean principles, and quality analysis, while also improving communication, collaboration, and leadership capabilities.

Overall, this internship has been a highly enriching experience that has shaped understanding of industrial operations and continuous improvement. The learnings will contribute significantly to future roles in manufacturing, industrial engineering, or operations management.

A. Annexures

A.1 Sample Hourly Monitoring Sheet

figures/hourly_sheet_sample.png

Figure A.1: Hourly monitoring sheet (sample)

A.2 PSS and OPL Templates

A.3 Additional Figures and Charts